ARAB REPUBLIC OF EGYPT
MINISTRY OF EDUCATION & TECHNICAL EDUCATION
CENTRAL ADMINISTRATION FOR CURRICULUM DEVELOPMENT (CCIMD)

OFFICIAL SEAL
MOE APPROVED

PURE MATHEMATICS: ALGEBRA & SOLID GEOMETRY

Official Ministry of Education Textbook - Grade 12

ACADEMIC YEAR: 2025 / 2026 | GRADE 12 (GENERAL SECONDARY CERTIFICATE)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

The authorized textbook for Pure Mathematics (Algebra and 3D Solid Geometry) issued by the Egyptian Ministry of Education, covering combinatorial identities, polar/exponential complex forms, Gaussian elimination, and spatial analytical geometry.

CORE CURRICULUM HIGHLIGHTS:

- Complete Binomial Theorem and ratio properties
- Cube roots of unity (Omega) and trigonometric De Moivre identities
- Cramer rule, matrix rank, and matrix inversion techniques
- Equations of lines, planes, and shortest skew-line distances in space

APPROVED CHAPTER MODULES:

OFFICIAL PUBLICATION METADATA & VERIFICATION

Book Code: TH-ALG-SOL-G12
Pages: 224 Official Pages | Size: 18.4 MB
Publisher: Ministry of Education & Technical Education - Arab Republic of Egypt
Legal Deposit: 2025/14892 - ISBN: 978-977-12-2436-19
Verification URL: https://moe.gov.eg/ar/elearning-content/

TH-ALG-SOL-G

MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

The Pure Mathematics: Algebra & Solid Geometry curriculum for Grade 12 develops rigorous analytic and algebraic reasoning. It unites discrete combinatorics, complex analysis in polar and exponential forms, linear systems with matrix algebra, and three-dimensional spatial Euclidean geometry. This syllabus equips students for higher engineering, physics, and computer science degrees.

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30%	40%	30%
Knowledge & Recall	**Understanding & Application**	**Higher Order Thinking (HOTS)**
Definitions, laws & formulas	Calculations, proofs & models	Analysis, synthesis & non-routine

3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal: https://moe.gov.eg/ar/elearning-content/. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 120 Minutes (2 Hours) | Total Paper Marks: 30 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Permutations, Combinations & Binomial Theorem	5 wks	7 pts	23.3%	Combinatorics, Binomial Expansion, General Term
Complex Numbers & De Moivre Theorem	5 wks	7 pts	23.3%	Argand Plane, Polar/Euler Form, nth Roots of Unity
Matrices, Determinants & Linear Systems	4 wks	6 pts	20.0%	Rank, Inverse Matrix, Cramer Rule, Gaussian
3D Cartesian Coordinates & Vectors	4 wks	5 pts	16.7%	Dot/Cross Product, Sphere Equation, Direction Cosines
Lines and Planes in 3D Space	4 wks	5 pts	16.7%	Vector/Parametric/Cartesian Equations, Angles, Distances

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism)

Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.

Engineering Sciences (Civil, Mechanical, Electrical)

Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.

Computer Science & Algorithmics

Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: CHAPTER 1: PERMUTATIONS, COMBINATIONS & BINOMIAL THE

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Master fundamental counting principle with and without replacement, ordered and unordered collections.
- [+] Derive and manipulate permutations nPr and combinations nCr identities: nCr = nC(n-r) and nCr + nC(r-1) = (n+1)Cr.
- [+] Expand (a + b)^n using Binomial Theorem, identify the general term T_(r+1) = nCr * a^(n-r) * b^r, find middle terms and greatest coefficient.
- [+] Solve advanced systems involving ratios of consecutive terms: T_(r+1) / T_r = ((n - r + 1) / r) * (b / a).

B. Statutory Theorems & Analytical Principles

Fundamental Counting Principle: If task A has m ways and task B has n ways, sequential completion has m * n ways.

Binomial Theorem: (a + b)^n = Sum_{r=0}^n [nCr * a^(n-r) * b^r] for positive integer n.

Pascal Combinatorial Recurrence: nCr + nC(r-1) = (n+1)Cr.

Consecutive Term Ratio Theorem: T_(r+1) / T_r = ((n - r + 1) / r) * (Term_2 / Term_1).

C. Standard Mathematical Expressions & Operational Formulas

- `nPr = n! / (n - r)!`
- `nCr = n! / [ r! (n - r)! ] = nPr / r!`
- `nCr / nC(r-1) = (n - r + 1) / r`
- `T_(r+1) = nCr * (x)^(n-r) * (y)^r in expansion of (x + y)^n`
- `Sum of coefficients in (a x + b)^n = (a + b)^n [Substitute x = 1]`

D. National Examination Pitfalls & Common Student Errors

[!] Confusing order significance: using permutations where combinations are needed in geometric configurations (diagonals, triangles).

[!] Neglecting negative signs inside terms when computing T_(r+1) in expansions like (2x - 3/x)^n.

[!] Failing to verify integer constraints on r when seeking the term independent of x.

CHAPTER SPECIFICATION: CHAPTER 2: COMPLEX NUMBERS & DE MOIVRE THEOREM

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Represent complex numbers in Cartesian z = x + i y, trigonometric/polar z = r (cos theta + i sin theta), and Euler exponential z = r e^(i theta) forms.
- [+] Apply De Moivre theorem for integer and rational exponents: [r (cos theta + i sin theta)]^n = r^n (cos n theta + i sin n theta).
- [+] Extract all n-th roots of a complex number evenly distributed on the circle of radius r^(1/n).
- [+] Operate with the cubic roots of unity 1, omega, omega^2 and their algebraic properties (1 + omega + omega^2 = 0, omega^3 = 1).

B. Statutory Theorems & Analytical Principles

De Moivre Theorem: (cos theta + i sin theta)^n = cos(n theta) + i sin(n theta) for all rational n.

Euler Identity: e^(i theta) = cos theta + i sin theta, e^(i pi) + 1 = 0.

Roots of Unity Theorem: The n roots of 1 form vertices of a regular n-gon centered at the origin on the unit circle.

Properties of Omega: omega^2 + omega + 1 = 0, omega - omega^2 = +- i * sqrt(3), omega^(3k) = 1.

C. Standard Mathematical Expressions & Operational Formulas

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> Modulus: r = |z| = sqrt(x^2 + y^2), Principal argument: -pi < theta <= pi
> Multiplication: z1 * z2 = r1 r2 [cos(theta1 + theta2) + i sin(theta1 + theta2)]
> Division: z1 / z2 = (r1 / r2) [cos(theta1 - theta2) + i sin(theta1 - theta2)]
> Roots: z^(1/n) = r^(1/n) [cos((theta + 2 k pi)/n) + i sin((theta + 2 k pi)/n)], k = 0, 1, ..., n-1
> Omega identities: 1 + omega = -omega^2, 1 + omega^2 = -omega, omega + omega^2 = -1
```

D. National Examination Pitfalls & Common Student Errors

[!] Calculating arg(z) incorrectly in quadrants 2, 3, and 4 (must maintain -180 < theta <= 180 degrees).

[!] Missing the imaginary unit i in roots or confusing (omega - omega^2)^2 = -3.

[!] Applying De Moivre to non-standard trigonometric forms like (sin theta + i cos theta) without shifting by (pi/2 - theta).

CHAPTER SPECIFICATION: CHAPTER 3: MATRICES, DETERMINANTS & LINEAR SYSTEMS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Evaluate 2x2 and 3x3 determinants using cofactor expansion and Laplace development.
- [+] Apply triangular matrix determinant theorem and properties of row operations.
- [+] Find the adjugate matrix and multiplicative inverse A^(-1) = (1 / det(A)) * adj(A).
- [+] Determine matrix rank Rank(A) and augmented matrix rank Rank(A|B) to classify system solvability (Rouche-Capelli Theorem).

B. Statutory Theorems & Analytical Principles

Determinant Transpose Theorem: det(A^T) = det(A).

Product Theorem: det(A B) = det(A) * det(B); det(k A) = k^n det(A) for n x n matrix.

Invertibility Theorem: Square matrix A is invertible if and only if det(A) != 0.

Rouche-Capelli Theorem: Consistent system iff Rank(A) = Rank(A|B). Unique solution if rank = n; infinite solutions if rank < n.

C. Standard Mathematical Expressions & Operational Formulas

- > `Inverse matrix: A^(-1) = (1 / |A|) * Adj(A)`
- > `System solution: X = A^(-1) * B`
- > `Cramer Rule: x_i = det(A_i) / det(A)`
- > `Homogeneous system: Has non-trivial solutions iff det(A) = 0`

D. National Examination Pitfalls & Common Student Errors

[!] Confusing matrix multiplication properties: AB != BA in general.

[!] Sign errors when computing cofactor matrix (remember checkerboard pattern + - + -).

[!] Forgetting that row operations change determinant values (swapping rows multiplies det by -1).

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

COMBINATORICS & BINOMIAL THEOREM

$nPr = n! / (n - r)!$

$nCr = n! / [r! (n - r)!]$

$nCr = nC(n-r)$

$nCr + nC(r-1) = (n+1)Cr$

$nCr / nC(r-1) = (n - r + 1) / r$

$T_{(r+1)} = nCr * a^{(n-r)} * b^r$

$T_{(r+1)} / T_r = ((n - r + 1) / r) * (b / a)$

COMPLEX NUMBERS & DE MOIVRE

$z = x + i y = r (\cos\theta + i \sin\theta) = r e^{(i \theta)}$

$r = \sqrt{x^2 + y^2}, \tan\theta = y / x$

$z^n = r^n [\cos(n \theta) + i \sin(n \theta)]$

$z^{(1/n)} = r^{(1/n)} [\cos((\theta + 2k \pi)/n) + i \sin((\theta + 2k$

$\omega^3 = 1, 1 + \omega + \omega^2 = 0$

$\omega - \omega^2 = \pm i \sqrt{3}$

MATRICES & DETERMINANTS

$A^{(-1)} = (1 / |A|) * Adj(A)$

$|A^T| = |A|, |A B| = |A| |B|, |k A| = k^n |A|$

$X = A^{(-1)} B$ (Matrix Equation)

$Rank(A) \le \min(m, n)$

VECTORS & 3D ANALYTICAL GEOMETRY

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$u \cdot v = |u| |v| \cos\theta = u_1 v_1 + u_2 v_2 + u_3 v_3$

$u \times v = | i\ j\ k ; u_1\ u_2\ u_3 ; v_1\ v_2\ v_3 |$

$d = |a x_0 + b y_0 + c z_0 + d| / \sqrt{a^2 + b^2 + c^2}$

$\cos\theta = |n_1 \cdot n_2| / (|n_1| |n_2|)$

$\sin\theta = |d \cdot n| / (|d| |n|)$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	120 Minutes (2 Hours)
Total Scaled Marks:	30 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- Write complete step-by-step reasoning for all free-response questions.
- Show all matrix row operations explicitly when calculating determinants and inverses.
- Always specify the direction angles in degrees or radians with unambiguous notation.
- Verify that all complex arguments lie within the principal interval (-pi, pi].

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION

NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

`Official Document Code: TH-ALG-SOL-G12`

Curriculum Cycle: 2025/2026 Academic Year | National Edition

`Official Portal: https://moe.gov.eg/ar/elearning-content/`